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MILITARY SPECIFICATION

HEATER, IMMERSION, LIQUID FUEL FIRED FOR TANK TRAILERS

1. SCOPE

1.1 Scope. This specification covers a liquid fuel burning immersion type heater designed primarily to warm water in tank trailers in order to prevent freezing of contents.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

P-D-680	-	Dry Cleaning Solvent.
QQ-A-225/5	-	Aluminum Alloy Bar, Rod, and Wire; Rolled, Drawn, or Cold Finished, 2017.
QQ-B-621	-	Brass Castings, Leaded Yellow.
QQ-B-626	-	Brass, Leaded and Non-Leaded; Rod, Shapes, Forgings, and Flat Products With Finished Edges (Bar, Flat Wire, and Strip).
QQ-B-1005	-	Bronze Castings, Leaded Tin and High Leaded Tin.
QQ-L-201	-	Lead Sheet.
QQ-S-561	-	Solder; Silver.
QQ-S-630	-	Steel, Bar, Carbon, Hot Rolled, (Merchant Quality).

FSC 4540

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QQ-S-634 - Steel, Bar, Carbon, Cold Finished (Standard Quality).
QQ-S-741 - Steel, Carbon: Structural Shapes, Plates, and Bars.
QQ-S-766 - Steel Plates, Sheets, and Strip - Corrosion Resisting.
QQ-S-781 - Steel, Strapping, Flat.
QQ-W-461 - Wire, Steel, Carbon (Round, Bare and Coated).
QQ-Z-325 - Zinc Coating, Electrodeposited, Requirements for.
RR-C-271 - Chains and Attachments, Welded, Weldless, and Roller Chain.
TT-C-490 - Cleaning Methods and Pretreatment of Ferrous Surfaces for Organic Coating.
TT-E-529 - Enamel, Alkyd, Semi-Gloss.
TT-P-636 - Primer Coating, Alkyd, Wood and Ferrous Metal.
WW-P-406 - Pipe, Steel (Seamless and Welded) (for Ordinary Use).
WW-P-521 - Pipe Fittings, Flange Fittings and Flanges, Ferrous and Steel, (Screwed and Butt-Welded), 150 Pound.
PPP-B-585 - Boxes, Wood, Wirebound.
PPP-C-650 - Crates, Wood, Open and Covered.
PPP-T-60 - Tape, Pressure-Sensitive Adhesive, Waterproof, for Packaging.

MILITARY

MIL-B-121 - Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-P-514 - Plates, Identification, Instruction and Marking, Blank.
MIL-P-551 - Pipes and Elbows, Air Conditioning-Heating.
MIL-R-3065 - Rubber, Fabricated Products.
MIL-L-3150 - Lubricating Oil, Preservative, Medium.
MIL-C-10382 - Corrosion Preventive, Petrolatum, Spraying Application: For Food Handling Machinery and Equipment.
MIL-C-16173 - Corrosion Preventive Compound, Solvent Cutback, Cold-Application.
MIL-P-19834 - Plates, Identification, Metal Foil, Adhesive Backed.
MIL-L-21260 - Lubricating Oil, Internal Combustion Engine, Preservative.

STANDARDS

FEDERAL

FED-STD-595 - Colors.

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MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-417 - Rubber Compositions, Vulcanized, General Purpose, Solid (Symbol and Tests).

DRAWINGS

ARMY NATICK LABORATORIES

5-13-446 - Heater, Immersion, Liquid Fuel Fired, for Tank Trailer. Assembly, Details, and List of Pieces.
5-13-447 - Heater, Immersion, Liquid Fuel Fired, for Tank Trailer. Heater Body Assembly and Details.
5-13-448 - Heater, Immersion, Liquid Fuel Fired, for Tank Trailer. Heater Body Details.
5-13-449 - Heater, Immersion, Liquid Fuel Fired, for Tank Trailer. Fuel Tank Strap.
5-13-453 - Heater, Immersion, Liquid Fuel Fired. Parts Common to Heater for Tank Trailer and Can, Corrugated. Fuel Tank Strap Details.
5-13-454 - Heater, Immersion, Liquid Fuel Fired. Parts Common to Heater for Tank Trailer and Can, Corrugated. Collar and Hood, Assembly and Details.
5-13-455 - Heater, Immersion, Liquid Fuel Fired. Hood Common to Heater for Tank Trailer and Can, Corrugated.
5-13-456 - Heater, Immersion, Liquid Fuel Fired. Parts Common to Heater for Tank Trailer and Can, Corrugated. Tank, Filler Plug, and Valve Assemblies.
5-13-457 - Heater, Immersion, Liquid Fuel Fired. Filler Plug Details Common to Heater for Tank Trailer and Can, Corrugated.
5-13-458 - Heater, Immersion, Liquid Fuel Fired. Valve, Gasoline Common to Heater for Tank Trailer and Can, Corrugated.
5-13-459 - Heater, Immersion, Liquid Fuel Fired. Burner Common to Heater for Tank Trailer and Can, Corrugated.
5-13-460 - Heater, Immersion, Liquid Fuel Fired for Tank Trailer. Lighter Torch.
5-13-1876 - Heater, Immersion, Liquid Fuel Fired, for Tank Trailer: Container for Level "A" Shipment.

(Miniature copies of Drawings 5-13-446 and 5-13-1876, identified as figures 1 and 2, respectively, are attached for information purposes only.)

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(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

* 2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Welding Society Code

Standard Qualification Procedure
of the American Welding Society

(Application for copies of the American Welding Society Code should be addressed to the American Welding Society, 33 West 39th Street, New York, N.Y. 10018).

American Society of Mechanical Engineers' Code

Welding Qualification of the
American Society of Mechanical Engineers

(Application for copies of the ASME Codes should be addressed to the American Society of Mechanical Engineers, 29 West 39th Street, New York, N.Y. 10018).

National Motor Freight Traffic Association, Inc., Agent

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N. W., Washington, D. C. 20036.)

Uniform Classification Committee, Agent

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, Illinois 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

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3. REQUIREMENTS

* 3.1 First article. When specified (see 6.2), the supplier shall furnish a sample for first article inspection and approval (see 4.2, 6.2 and 6.3).

3.2 Materials. The materials specified shall conform to the applicable specifications and requirements specified hereinafter. All materials used in the fabrication of the heater shall be cleaned and free from any substance, such as dirt, grease, or oils that may be toxic.

3.2.1 Steel, carbon.

3.2.1.1 Sheet and strip. Carbon steel sheets and strips shall be commercial quality.

3.2.1.2 Bar stock. Carbon bar stock steel shall conform to QQ-S-630 or QQ-S-634.

3.2.1.3 Wire. Carbon steel wire shall conform to QQ-W-461.

3.2.1.4 Angles. Steel angles shall conform to type II of QQ-S-741.

3.2.2 Steel, corrosion-resisting, sheet and strip. Corrosion-resisting steel sheet and strip shall conform to chemical composition requirements of QQ-S-766.

3.2.3 Brass.

* 3.2.3.1 Castings. Brass used for the valve body casting shall conform to composition B of QQ-B-621 or bronze, conforming to composition 11 of QQ-B-1005.

3.2.3.2 Forgings. Brass used for forgings shall conform to composition 21 of QQ-B-626.

3.2.3.3 Rod. Brass rod shall be half-hard conforming to composition 22 of QQ-B-626.

3.2.4 Aluminum. Aluminum alloy 2017 shall conform to condition T4 of QQ-A-225/5.

3.2.5 Solder. Solder shall conform to QQ-S-561.

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3.2.6 Gaskets.

3.2.6.1 Filler plug gasket. The filler plug gasket shall be fabricated from sheet lead conforming to grade B of QQ-L-201.

3.2.6.2 Vent plug gasket. The vent plug gasket shall be of material conforming to type, class, grade, and suffix letters of SB615AE5 of MIL-R-3065 and MIL-STD-417.

3.2.7 Pipe, outlet. The pipe used in the fabrication of the outlet nipple shall be standard black pipe conforming to type I, class A of WW-P-406.

3.2.8 Elbow. The malleable iron elbow shall conform to type 1 of WW-P-521.

3.2.9 Chains.

3.2.9.1 Chain, safety. The safety chain shall be weldless chain, brass, approximately 27 links per foot conforming to type II, class 6 of RR-C-271.

3.2.9.2 Chain, bead pattern. The bead pattern chain shall be a commercial quality free swivel, bead pattern as shown on Drawing 5-13-446 and 5-13-453.

3.2.10 Asbestos.

3.2.10.1 Wick burner. The asbestos for burner wick shall be soft, highly absorbent, long fiber, pure asbestos roll board.

3.2.10.2 Valve packing. The asbestos for the valve packing shall be asbestos rope impregnated with graphite and shall be made from corded long-fiber asbestos, spun into yarn and laid up into the general form of a loosely twisted cord. If a core is employed, it shall be of the same material and construction.

3.2.10.3 Burner lighter. The asbestos used for the burner lighter shall be soft, highly absorbent long fiber in the form of braided tubing.

3.2.11 Hardware. All hardware indicated by an asterisk on Drawing 5-13-446 shall be of commercial quality conforming to the material and finish as specified on drawings.

3.2.12 Pipe, stove. The stove pipe shall be 4-inch diameter and conforming to MIL-P-551. Four lengths shall be furnished with each heater.

3.2.13 Primer. Primer shall conform to TT-P-636.

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3.2.14 Enamel. Enamel shall conform to class A of TT-E-529. The color shall be olive drab to approximate color No. X24087 of Fed. Std. No. 595.

3.3 Construction. The construction of all components and assemblies of the items covered herein shall conform to the requirements of this specification in the design, details, dimensions, and material classifications and size thereof shown on drawings.

3.3.1 Welders. The welders performing the welding shall be certified welders and have passed the qualification test as prescribed by either the Standard Qualification Procedures of the American Welding Society or Welding Qualification of the American Society of Mechanical Engineers' code. The supplier will furnish the Government with a list of names of his welders who are certified and have passed the test of either referenced codes and he shall certify that the welding of this equipment was performed by the welders listed.

3.4 Details of components.

3.4.1 Body. The heater body assembly shown on Drawing 5-13-447 shall not leak when tested in accordance with 4.4.1.

3.4.2 Fuel tanks. The fuel tank assembly shown on Drawing 5-13-456 shall not leak when tested in accordance with 4.4.2.

* 3.4.3 Burner plate. The burner plate assembly holes through the top plate, wick and bottom plate, as shown on Drawing 5-13-459, shall be in alignment within plus or minus 0.010 inches.

3.4.4 Filler plug assembly. The filler plug and gasket shall conform to Drawing 5-13-457. The gasket shall be assembled on filler plug, with a press fit, to secure gasket to plug.

* 3.4.5 Vent plug and fuel valve. The vent plug and fuel valve shown on Drawings 5-13-449 and 5-13-456 shall have no internal obstruction when tested in accordance with 4.4.2.

3.5 Finish.

3.5.1 Galvanizing. Prior to painting, the filler plug assembly without gasket shall be given an electrogalvanized finish conforming to type II, class 2 of QQ-Z-325.

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3.5.2 Copper plating. Copper plating of the burner lighter metal components shall be in accordance with the manufacturer's standard practice.

* 3.5.3 Priming. Prior to painting the surfaces, as indicated, the following heater components shall be thoroughly cleaned by any method or type of TT-C-490 and given one complete coat of primer specified in 3.2.13.

a. Heater body - outside surface above the top component.

b. Top heater body - upper surface, and outside and inside surface of rim.

c. Fuel tank - all outside surfaces including filler plug, vent plug, and gasoline valve, while screwed shut and elbow and nipple screw in place.

d. Hood prior to assembly of instruction plate - all surfaces.

e. Draft gate - all surfaces.

f. Fuel tank strap assembly - all surfaces except male and female threads.

g. Hinge pin and lug pin with cotter pins - all surfaces.

h. Nuts, screwheads, rivet heads assembled in place - all outside surfaces.

Any primer application reaching within 3 inches inside of the burner space or flue will not be considered detrimental.

3.5.4 Enameling. When thoroughly dry, the primed surfaces specified in 3.5.3 shall be given a complete coat of enamel specified in 3.2.14. Any enameling application reaching within 3 inches inside the burner space or flue shall not be considered detrimental. Touchup of surfaces marred inadvertingly will be acceptable.

3.6 Marking for identification and instruction.

3.6.1 Burner and draft gate identification. Burner and draft gate identification shall be indented as shown on drawings 5-13-455 and 5-13-459 respectively. The indentation shall show no sign of penetration, unevenness or distortion on the back side of the burner leg or draft gate.

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3.6.2 Fuel tank identification and instructions. Fuel tank identification and instructions shown on Drawing 5-13-456 shall be printed on a black, non-specular surface, adhesive backed foil conforming to MIL-P-19834. Lettering shall be aluminum color.

3.6.3 Heater identification and instructions. Heater identification and instructions as shown on drawings 5-13-447 and 5-13-454 respectively shall be applied to type III, composition C plate conforming to MIL-P-514 (see 6.4).

3.6.4 Warning plate. A warning plate consisting of "Warning - This heater must not be used in insulated plastic tank trailers.", shall be printed in 1/4 inch block letters on aluminum photosensitized plate as specified in 3.6.3. The plate shall be located on the top of the heater body assembly opposite the 2 chain-pin and chain-lug assemblies and affixed with 2 No. 6 by 3/8 inch long cadmium plated steel drive screws.

3.7 Workmanship. All component parts of the heater assembly shall be clean, free of dents, distortions, burrs and slivers. The complete heater assembly shall be square and component parts shall be in alignment. Enameled surfaces shall be smooth, adherent, free of blisters, sags, runs and not tacky. The vent plug and fuel valve when opened shall have no paint obstruction to the flow of air when tested in accordance with 4.4.2.

3.7.1 Welding and soldering. All welds and soldering shall be clean, adherent, continuous, not burned through, free of cracks, pits, flux and weld splatter.

3.7.2 Rivets. All rivets shall be tight and peened with smooth and solid heads.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 First article. When a first article is required, it shall be examined in accordance with the defects in table I, dimensions specified and tested in accordance with 4.4.1 and 4.4.2. The presence of any visual defects, dimensions not specified or failure to pass any test shall be cause for rejection of the first article.

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4.3 Inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.3.1 Component and material inspection. In accordance with 4.1 above components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase documents.

* 4.3.2 In-process inspection.

4.3.2.1 Toxic material. Examination shall be made for existence of any toxic material to determine compliance with 3.2. When nonconformance is noted, correction shall be made to the affected items and process.

4.3.2.2 Galvanizing and priming. Examination shall be made to determine compliance with galvanizing and priming requirements of 3.5.1 and 3.5.3 respectively. When nonconformance is noted correction shall be made to the affected items and process.

* 4.3.2.3 Intermediate test. Each heater body and fuel tank shall be tested in accordance with 4.4.1 and 4.4.2. Evidence of failure to comply with specified requirements shall be cause to reject the nonconforming items.

4.3.3 End item inspection. The lot shall consist of all immersion heater units offered for inspection at one time. The sample unit shall be one completely assembled heater.

4.3.3.1 Visual examination. Examination shall be made of the end item for defects listed in table I. The inspection level shall be S-2 and the acceptable quality level (AQL) shall be 2.5 for major and 6.5 for total defects expressed as defects per 100 units.

TABLE I. Classification of defects

Examine	Defect	Classification Major	Minor
Finish	Not finished.	X	
	Type or color of finish not as specified or blistered, peeled, chipped, or area of no film.		X
	Poor touchup, color separation, stained, overspray, or foreign matter imbedded in finish; or finish otherwise impaired.		X

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TABLE I. Classification of defects (cont'd)

Examine	Defect	Classification Major	Minor
Materials	Materials not as specified.	X	
Construction and workmanship, general (applicable to all components, subassemblies and assemblies)	Any part missing.	X	
	Fractured, split, punctured, torn, dented, creased, deteriorated, bowed, sprung, malformed or miscast.	X	
	Component missing, misplaced, or not in proper alignment.	X	
	Component loose.		X
	Evidence of sharp burrs, slivers, or edges.	X	
	Any functioning component that will not operate as intended.	X	
	Any functioning component that requires abnormal force to operate.		X
	Not connected or joined as specified or poorly accomplished.	X	
Welding and brazing (where applicable)	Not type specified, missing, incomplete, through burn holes, fractured, not fused or adherent.	X	
	Slag inclusion, slight undercut, not smooth and uniform, or deposits of scale and flux not removed.		X
Soldering	Missing or not adherent	X	
	Incomplete	X	
	Not clean (flux or flux residue not removed), pinholes in solder		X
Bolts, nuts, screws, studs and other threaded fastener components	Missing, not type specified, broken, stripped, or fractured	X	
	Loose		X
Rivets (when applicable)	Missing, not approved type, loose, broken, not peened	X	
Pipe, tubing, fittings and valves	Improperly connected or installed, will not provide a seam where required, or stripped	X	

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TABLE I. Classification of defects (cont'd)

Examine	Defect	Classification Major	Classification Minor
Cleanliness of unit	Not clean		X
Lighter	Missing	X	
Assembly	Any component not properly assembled or secured	X	
	Unit perceptibly out of square or alignment	X	
	Any assembly or component that does not provide a seal where required	X	
Marking, identification, instructions, warning plate, manuals, nameplate, parts list, et cetera	Missing, incomplete, or illegible, not specified type or size, not in proper location, not affixed in the prescribed manner, not correct color, not stamped, embossed, printed, vitreous enameled, as specified		X

4.3.3.2 Dimensional examination. Inspection shall be made of the end item to determine compliance with dimensions specified. Any deviation from specified requirements shall constitute a defect. The inspection level shall be S-2 and the AQL shall be 4.0 defects expressed in terms of defects per 100 units.

4.3.3.3 Examination of preparation for delivery. An examination shall be made to determine that preservation, packaging, packing and marking as required by section 5 of this specification are complied with. Defects will be as indicated in table II. The sample unit shall be one shipping container fully prepared for delivery. The lot shall be the number of containers offered at one time. The inspection level shall be S-2 with an AQL of 4.0 defects expressed in terms of defects per 100 units.

TABLE II. Preservation and packaging defects

Examine	Defect
Markings, interior and exterior	Omitted, incorrect, illegible, of improper size, location, sequence or method of application.

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TABLE II. Preservation and packaging defects (cont'd)

Examine	Defect
Preservation	Preservative improperly applied or missing.
Materials	Component missing, damaged, or otherwise defective.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling; bulging or distortion of containers, blocking and cushioning inadequate, improper, or missing.

4.3.3.3.1 Examination of shipping containers. When shipping containers are required to be in accordance with PPP-B-585 or PPP-C-650, examination for defects in closure and strapping shall be in accordance with the applicable container specification.

4.4 Tests.

4.4.1 Body heater test. The heater body shall be submerged in water and filled with compressed air at a pressure of between 4 and 5 pounds per square inch gage and checked for leaks to determine compliance with 3.4.1.

4.4.2 Fuel tank. The fuel tank with vent plug and fuel valve closed shall be submerged in water and filled with compressed air through the filler hole at a pressure of between 4 and 5 pounds per square inch gage and checked for leaks to determine compliance with 3.4.2. After testing for leaks the vent plug and valve shall be opened and checked for flow of air to determine compliance with 3.4.5 and 3.7.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Disassembly. Each heater shall have the fuel tank and fuel tank strap detached from the body of the heater and the cylindrical nuts replaced and finger tightened. The four stovepipe joints shall be detached from the heater and disassembled. The burner shall be removed.

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5.1.1.2 Cleaning. All uncoated ferrous metal surfaces of the heater and detached components shall be cleaned with dry-cleaning solvent conforming to P-D-680 and thoroughly dried prior to application of preservative.

* 5.1.1.3 Preservation. The interior of the fuel tank and interior of the body of the heater shall be fogged with preservative conforming to grade 1 of MIL-L-21260 and all excess preservative drained off. All exterior uncoated surfaces of the heater body and fuel tank valve stem shall be coated with preservative conforming to MIL-C-10382. The lighter rod, ring and staples shall be coated with preservative conforming to grade 1 of MIL-C-16173. The burner shall be lightly coated with preservative conforming to MIL-L-3150 and wrapped in one thickness of greaseproof barrier-material conforming to type II, grade A of MIL-B-121, with the wrap secured by tape conforming to type IV class of PPP-T-60. The four cleaned stovepipe joints shall be individually coated with medium preservative oil conforming to MIL-L-3150.

5.1.1.4 Packaging.

5.1.1.4.1 Heater body. The burner shall be suspended and secured within the burner flue with minimum .040 inch diameter annealed wire. The hood shall be sealed in place over the burner flue with tape as specified herein. The preserved lighter shall be placed within the exhaust flue. The draft gate shall be sealed in place with tape as specified herein.

5.1.1.4.2 Fuel tank. The outlet pipe and valve assembly of the fuel tank shall be wrapped with greaseproof barrier-material secured in place with tape as specified in 5.1.1.3. The fuel tank strap shall be mounted on the fuel tank and nuts tightened.

5.1.1.4.3 Stovepipe. The rivets used in the stovepipe joints as retaining means shall be placed in a cotton cloth drawstring bag and securely tied to the outer stovepipe through one of the rivet holes. The preserved stovepipe joints shall be nested one within another and wrapped with greaseproof barrier-material as specified in 5.1.1.3. The wrap shall completely inclose the stove pipe including the ends and shall be secured in place with tape as specified in 5.1.1.3.

5.1.1.4.4 Placement. The fuel tank, with tank strap attached, shall be placed on end on top of the lower portion of the body of the heater, centered against the flue stack and the vertical and diagonal braces, and securely held in place by two 3/8 by 0.015-inch flat steel straps conforming to class B of QQ-S-781. The straps shall pass around the tank and the flue stack and shall be spaced not less than 6 inches apart. Care shall be taken to position the outlet pipe and valve assembly of the tank so that it will not come in contact with the heater or the shipping container. The packaged stovepipe shall be placed upright beside the vertical brace within the flanged edge of the heater top and securely wired to the vertical brace.

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5.1.2 Level C. Heaters shall be preserved and packaged to afford adequate protection against corrosion, deterioration and damage during shipment from the supply source to the first receiving activity. The supplier may use his standard practice when it meets these requirements.

5.2 Packing. Packing shall be level A or C as specified (see 6.2).

5.2.1 Level A. Each heater, complete with components, preserved and packaged as specified in 5.1, shall be packed in a nailed wood crate conforming to PPP-C-650 and the nailed wood container illustration shown on Drawing 5-13-1876; or a wirebound container conforming to PPP-B-585 and the wirebound container illustration shown on Drawing 5-13-1876. Each heater shall be positioned so that the packaged stovepipe is in one corner of the shipping container. Closure and strapping shall be in accordance with the appendix of the applicable container specification.

* 5.2.2 Level C. Heaters, preserved and packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with Uniform Freight Classification Rules or National Motor Freight Classification Rules as applicable.

5.3 Marking. In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The immersion water heater is intended for insertion in a steel or aluminum water tank trailer for the purpose of heating water to prevent freezing and is not intended for use in the insulated water tank trailer. The heater is designed to use standard leaded motor gasoline. It will also operate satisfactorily using any other grade of gasoline that may be available. It may be used with kerosene or light diesel oil as fuel; however, its operation will be below par.

* 6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) When a first article is required (see 3.1, 4.2 and 6.3).
(c) Selection of applicable levels of preservation, packaging, and packing (see 5.1 and 5.2).

* 6.2.1 Contract data requirements. Any requirement for equipment manuals for the items covered by this specification should be included in DD Form 1423 Contract Data Requirements List and cited in the contract.

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* 6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of ASPR 7-104.55. The first article should be a preproduction sample, initial production item or other specific item described under the definition of a First Article in the ASPR. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for inspection and approval of the first article.

6.4 Photographic sensitized plate is available through the General Services Administration Federal Supply Schedule under FSC class 6750, photographic sensitized and chemical supplies. Photographic sensitized plate made by Metal Photo of Cincinnati, Ohio or equal is satisfactory (see 3.6.3).

* 6.5 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodian:

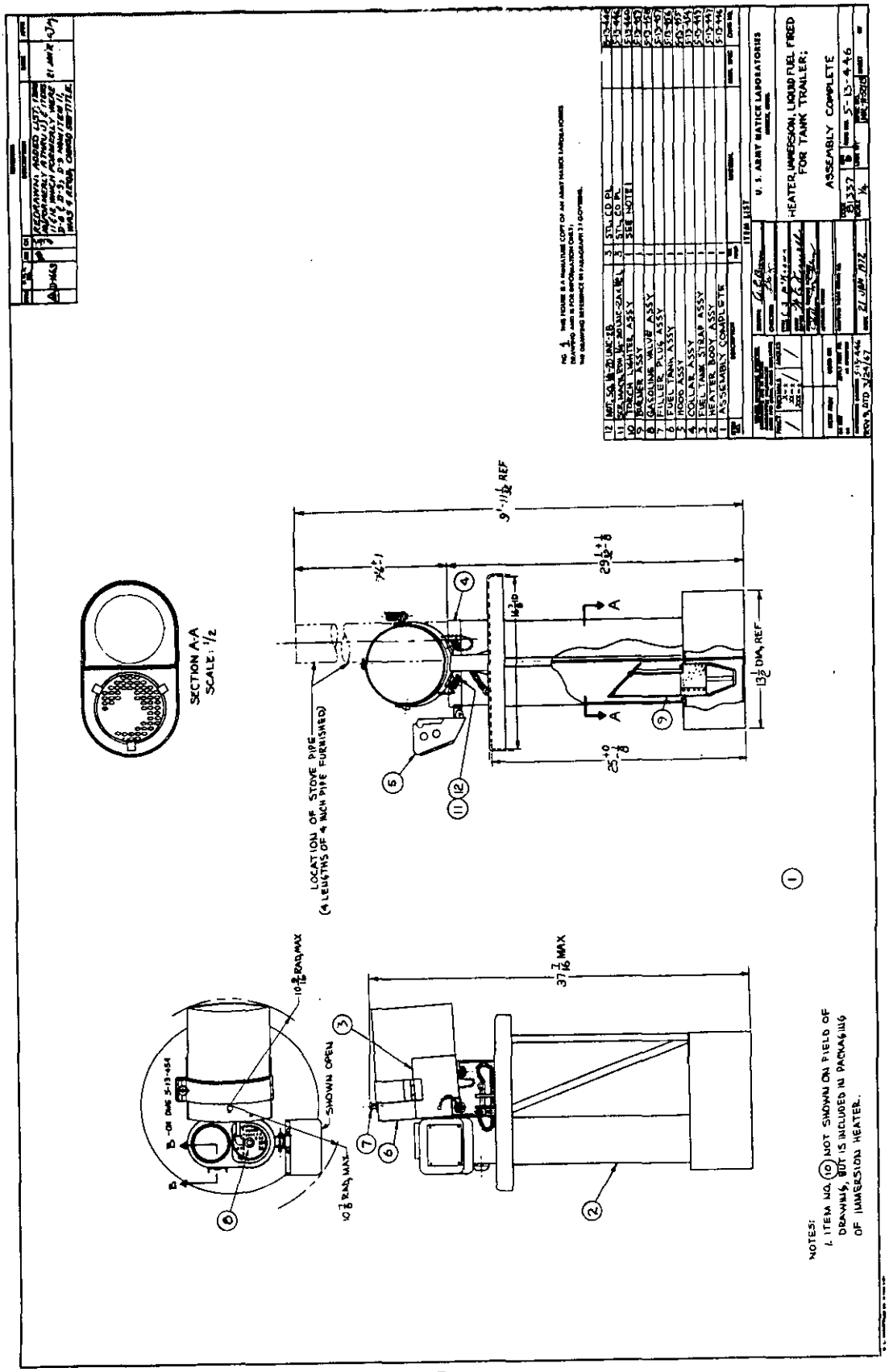
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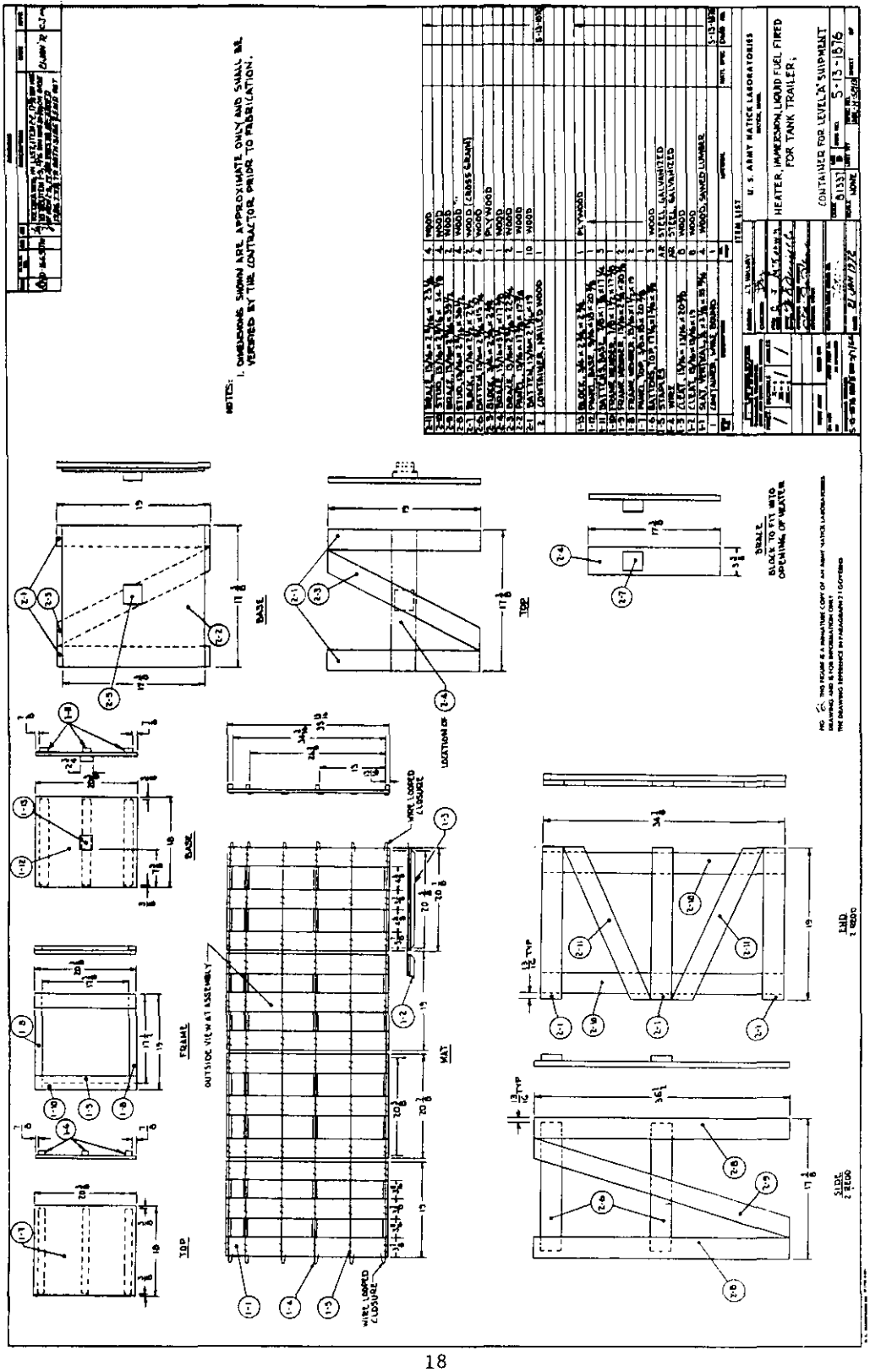
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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

Heater, Immersion, Liquid Fuel Fired For Tank Trailers MIL-H-52113B(GL)

ORGANIZATION

CITY AND STATE	CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*	DATE

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